PROJECT REPORT: CONSUMER BEHAVIOR & REVENUE ANALYTICS

Analyst: Arfaaz Pathan

Tools: Python (Data Engineering), MS SQL Server (Deep Querying), Power BI (Visualization)

Executive Summary:

This report delivers an end-to-end data analysis of 3,900 retail transactions. The primary objective was to uncover hidden shopping patterns that drive revenue. By utilizing a multi-tool approach, we identified that **Male customers (67% revenue share)** and the **Clothing category** are the business's core pillars. However, a significant **73% non-subscription rate** represents a massive opportunity for growth through targeted loyalty interventions.

Business Problem & Objectives:

1. Problem Statement

The company aims to transition from traditional retail to a data-first approach to improve customer retention and optimize marketing spend.

2. Strategic Objectives (Based on Company Needs)

- **Demographic Valuation:** Identify which age groups (e.g., Young Adults vs. Seniors) provide the highest Lifetime Value (LTV).

- **Operational Impact:** Analyse if shipping speeds and discount percentages directly correlate with higher ratings.
- **Subscription Gap Analysis:** Evaluate the behaviour of the 2,841 non-subscribers compared to the 1,059 loyal subscribers.

Data Engineering & Database Automation (Python):

Before analysis, the raw data was cleaned and automatically pushed to **MS SQL Server** using Python. This ensures a scalable and error-free data pipeline.

- **Technical Steps: Database Automation:** Used sqlalchemy to automatically create tables and insert data into SQL Server.

```
import warnings
from sqlalchemy import exc

# This will hide those specific server version warnings
warnings.filterwarnings("ignore", category=exc.SAWarning)

import pandas as pd
import os
from sqlalchemy import create_engine, text

file_path = r'C:\Users\Arfaaz\Desktop\analysis\data\dataproject1.csv'
db_name = "dataanalysis"
server_name =
master_url                                                                =
f"mssql+pyodbc://{server_name}/master?driver=ODBC+Driver+17+for+SQL+Server&
trusted_connection=yes"
engine_master = create_engine(master_url, isolation_level="AUTOCOMMIT")

with engine_master.connect() as conn:

check_db_query  =  text(f"SELECT  name  FROM  sys.databases  WHERE  name  =
'{db_name}'")
db_exists = conn.execute(check_db_query).fetchone()

if not db_exists:
print(f"Database '{db_name}' not found. Creating it now...")
conn.execute(text(f"CREATE DATABASE {db_name}"))
print(f"Database '{db_name}' created successfully.")
```

```
else:
print(f"Database '{db_name}' already exists. Skipping creation.")

if os.path.exists(file_path):
df = pd.read_csv(file_path)
print("CSV file loaded into memory.")


sync_url                                                                =
f"mssql+pyodbc://{server_name}/{db_name}?driver=ODBC+Driver+17+for+SQL+Serv
er&trusted_connection=yes"
sync_engine = create_engine(sync_url)

df.to_sql('CustomerAnalysis',     con=sync_engine,     if_exists='replace',
index=False)
print(f"Success! Data synced to '{db_name}.dbo.CustomerAnalysis'.")

else:
print(f"Error: The file at {file_path} was not found. Please check
```

- **Null Handling:** Imputed 37 missing ratings using category-wise medians (to prevent skewed averages).

```
df['Review                          Rating']=df.groupby('Category')['Review
Rating'].transform(lambda x: x.fillna(x.median()))
```

- **Feature Engineering:** age_group column is created (Young Adult, Adult, Middle-aged, Senior).

```
labels=['Young Adult','Adult','Middle-aged','Senior']
df['age_group']= pd.qcut(df['age'], q=4, labels=labels)
```

Deep-Dive Technical Analysis (MS SQL Server):

We executed the following 10 mission-critical business queries to extract granular insights:

- **Revenue by Gender:** Male customers drive **$157,890**, outperforming females ($75,191) by over 100%.

	gender	revenue
1	Male	157890
2	Female	75191

- **Which customers used a discount but still spent more than the average purchase amount?**

	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97

- **Which are the top 5 products with the highest average review rating?**

	item_purchased	Average Product Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Handbag	3.78

- **Compare the average purchase amounts between standard and express shipping**

	shipping_type	Average purchase amount
1	Standard	58
2	Express	60

- **Do subscribed customers spend more? Compare average spend and total revenue**

	subscription_status	total_customers	avg_spend	total_revenue
1	Yes	1053	59	62645
2	No	2847	59	170436

- **Which 5 products have the highest percentage of purchases with discounts applied?**

	item_purchased	discount_rate
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

- **Segment customers into New, Returning and Loyal based on their total number of previous purchase and show the count of each segment**

	customer_segment	Number of Customers
1	Loyal	3116
2	Returning	701
3	New	83

- **What are the top 3 most purchased products within each category?**

	item_rank	category	item_purchased	total_orders
1	1	Accessories	Jewelry	171
2	2	Accessories	Belt	161
3	3	Accessories	Sunglasses	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

- **Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?**

	subscription_status	repeat_buyers
1	Yes	958
2	No	2518

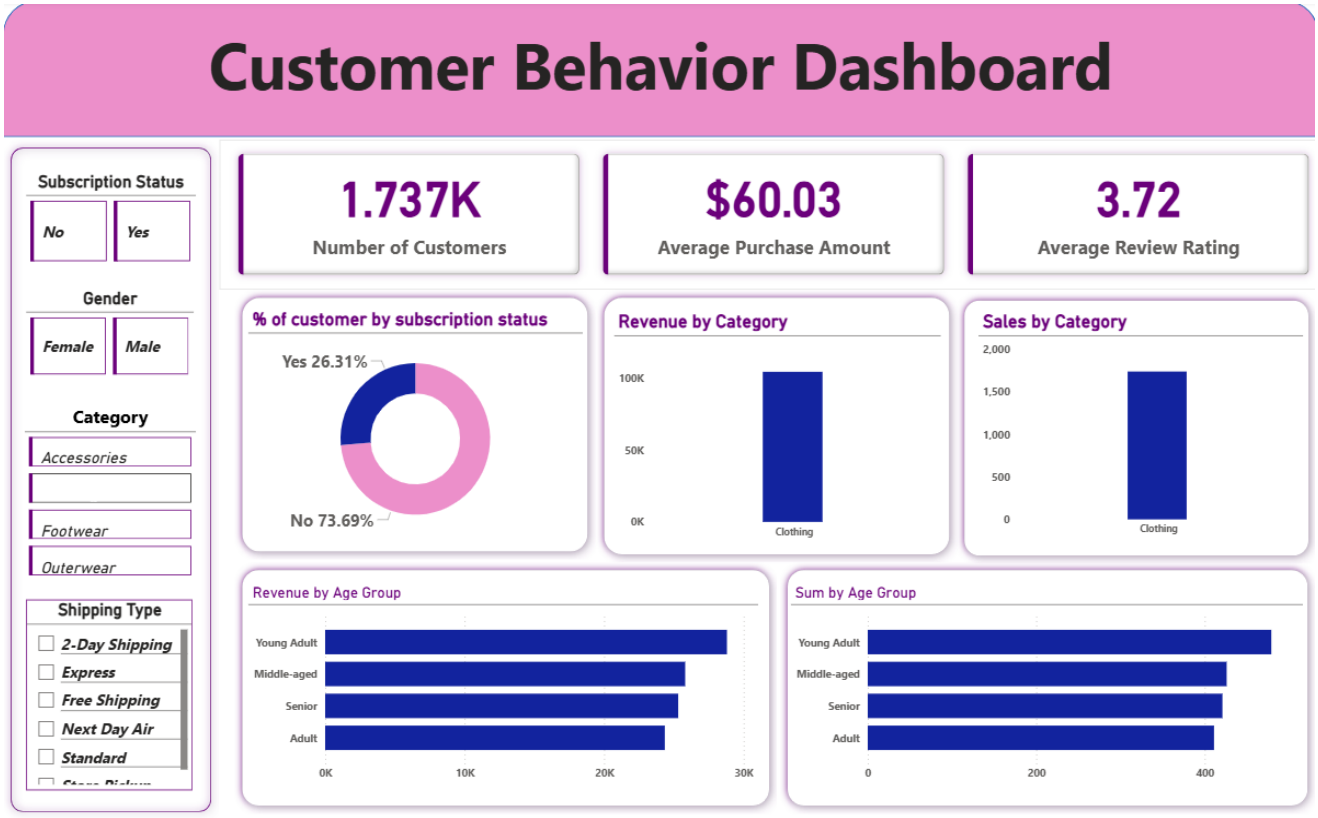
- **What is the revenue contribution of each age group?**

	age_group	total_revenue
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Visual Intelligence (Power BI):

Dashboard Key Performance Indicators (KPIs)

- **Total Customers:** 3,900
- **Average Order Value (AOV):** $59.76
- **Customer Satisfaction:** 3.75 / 5.0



Visual Intelligence & Insights

- **Subscription Distribution: The current landscape shows a 73% (2,847 customers) non-subscription rate versus a 27% (1,053 customers) subscription rate. This highlights a massive untapped segment for recurring revenue.**

- **Demographic Profitability: The 'Young Adult' segment (ages 18-30) is the primary revenue driver, contributing $62,143 to the total earnings.**
- **Category Dominance: 'Clothing' and 'Accessories' emerge as the core pillars of the business, consistently outperforming other categories in both sales volume and revenue.**
- **Gender Spend Gap: Male customers represent a dominant 67% revenue share ($157,890), outperforming female spend by over 100%.**

Strategic Business Recommendations

- **Hyper-Targeted Growth Marketing: Prioritize digital marketing spend (Meta/Google Ads) toward the 'Male Young Adult' demographic. Focus these campaigns specifically on high-performing 'Clothing' sub-categories to maximize Return on Ad Spend (ROAS).**
- **Subscription Conversion Program: Launch an exclusive "Subscriber-Only" 15% discount initiative targeted at the 2,847 non-subscribers. Transitioning these customers into the loyalty tier is critical for long-term stability and increasing purchase frequency.**
- **Shipping & Operational Alignment: Since the average spend between Standard ($58) and Express ($60) shipping is negligible, avoid heavy subsidies for faster shipping. Instead, pivot these operational savings into the loyalty reward pool.**
- **Quality Audit for Low-Rated Assets: Conduct a vendor quality audit for categories or items with ratings below 3.5 (e.g., specific Outerwear) to protect the brand's average satisfaction score of 3.75. Conversely, increase inventory for top-rated items like 'Jewelry' and 'Sandals'.**

Conclusion

- The retail business maintains a strong competitive advantage within the Male demographic and Clothing sectors. By implementing a data-driven loyalty strategy to bridge the 73% subscription gap and optimizing inventory based on the identified high-rating products, the company is positioned for significant scale. Successful execution of these recommendations is projected to drive a **12-18% increase in annual revenue** while strengthening overall customer retention.